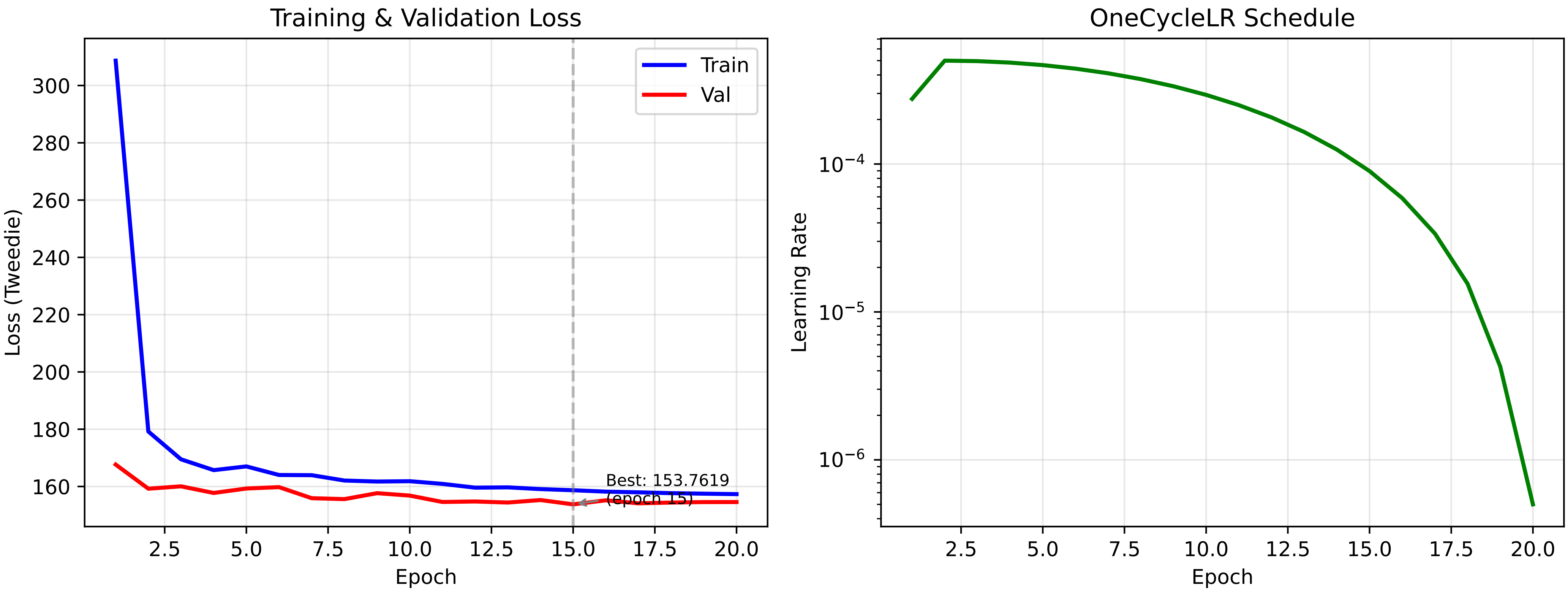
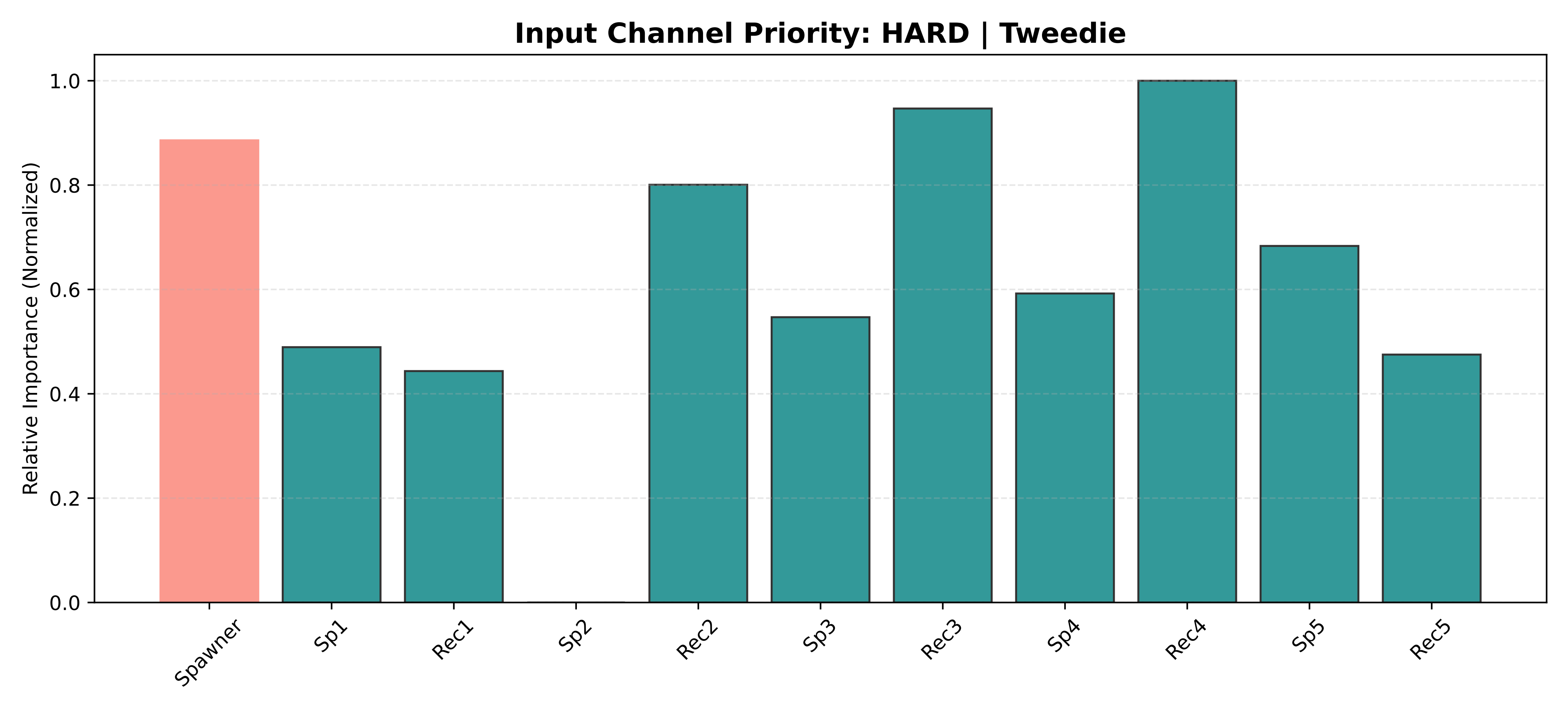
CrabTransformer Training Report

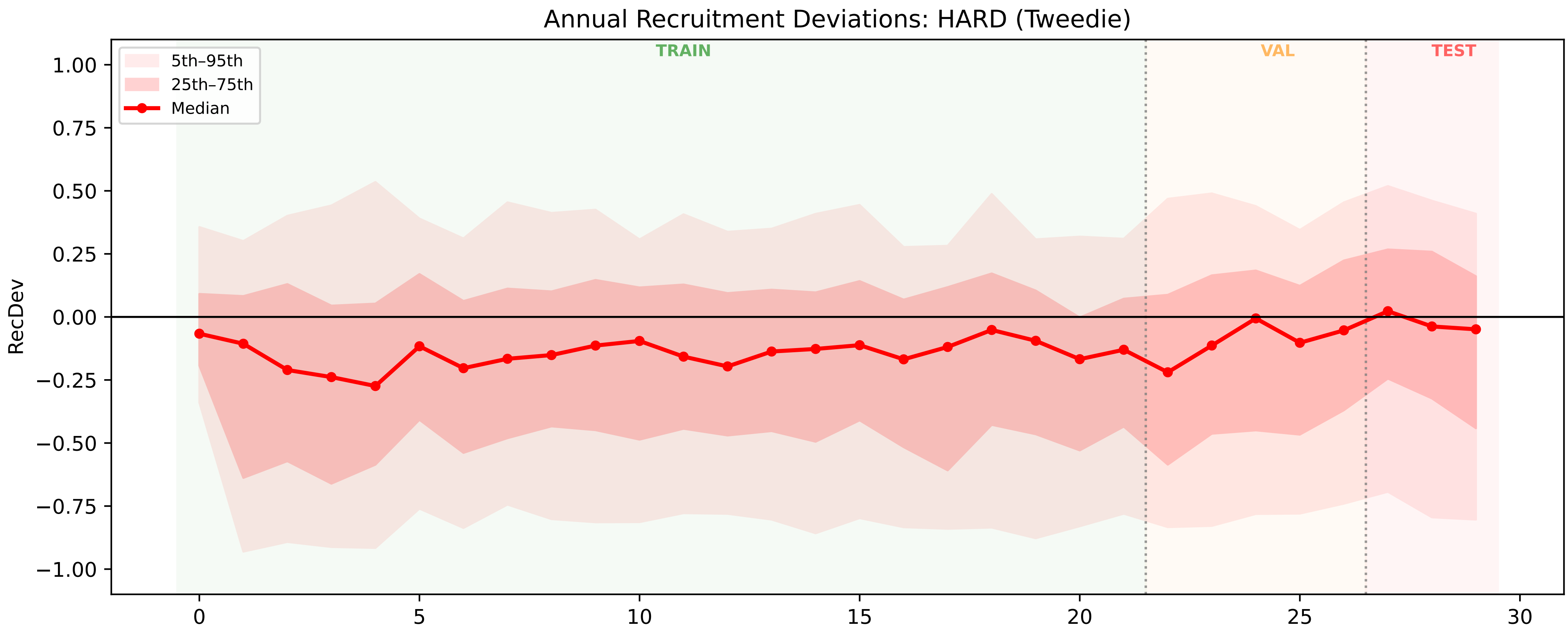
Scenario: HARD | Loss: Tweedie

```
─── Data Generation Parameters ───
  grid_size: 50     n_years: 30     n_bootstraps: 100
  mean_density: 3728.0     mean_recruit_density: 4514.0
  recruitment_correlation: 0.8     lag: 5
  spatial_kappa: 0.3     temporal_rho: 0.7

─── Training Configuration ───
  epochs_run: 20     best_val_epoch: 15
  best_val_loss: 153.7619     final_train_loss: 157.3361
  bias_correction (MSE): 1.0000

─── Model Architecture ───
  embed_dim: 64     num_heads: 4     num_layers: 3
  d_ff: 256     dropout: 0.2     patch_size: 5
  train/val/test years: 22/5/3
```

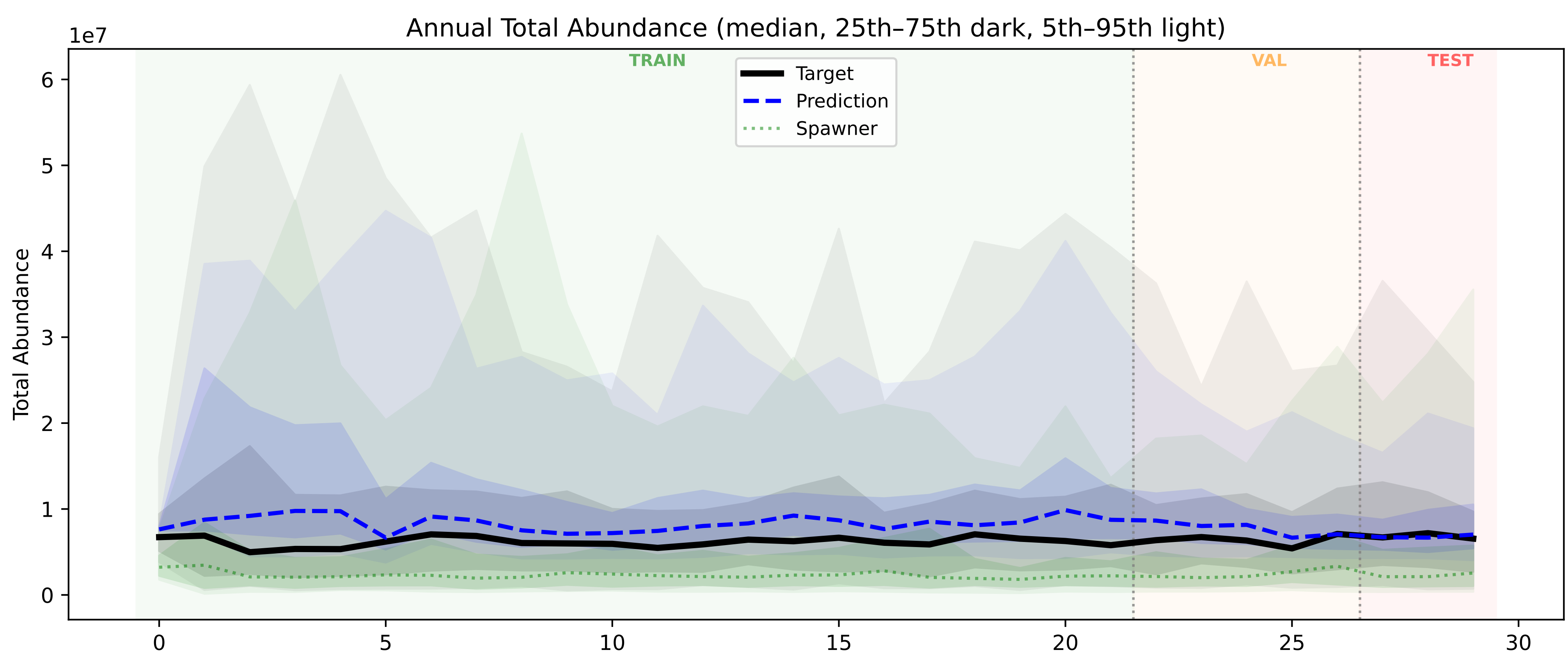
Training Curves: HARD | Tweedie




Input Channel Priority: HARD | Tweedie
Relative Importance (Normalized)
0.0
0.2
0.4
0.6
0.8
1.0
Spawner
Sp1
Rec1
Sp2
Rec2
Sp3
Rec3
Sp4
Rec4
Sp5
Rec5

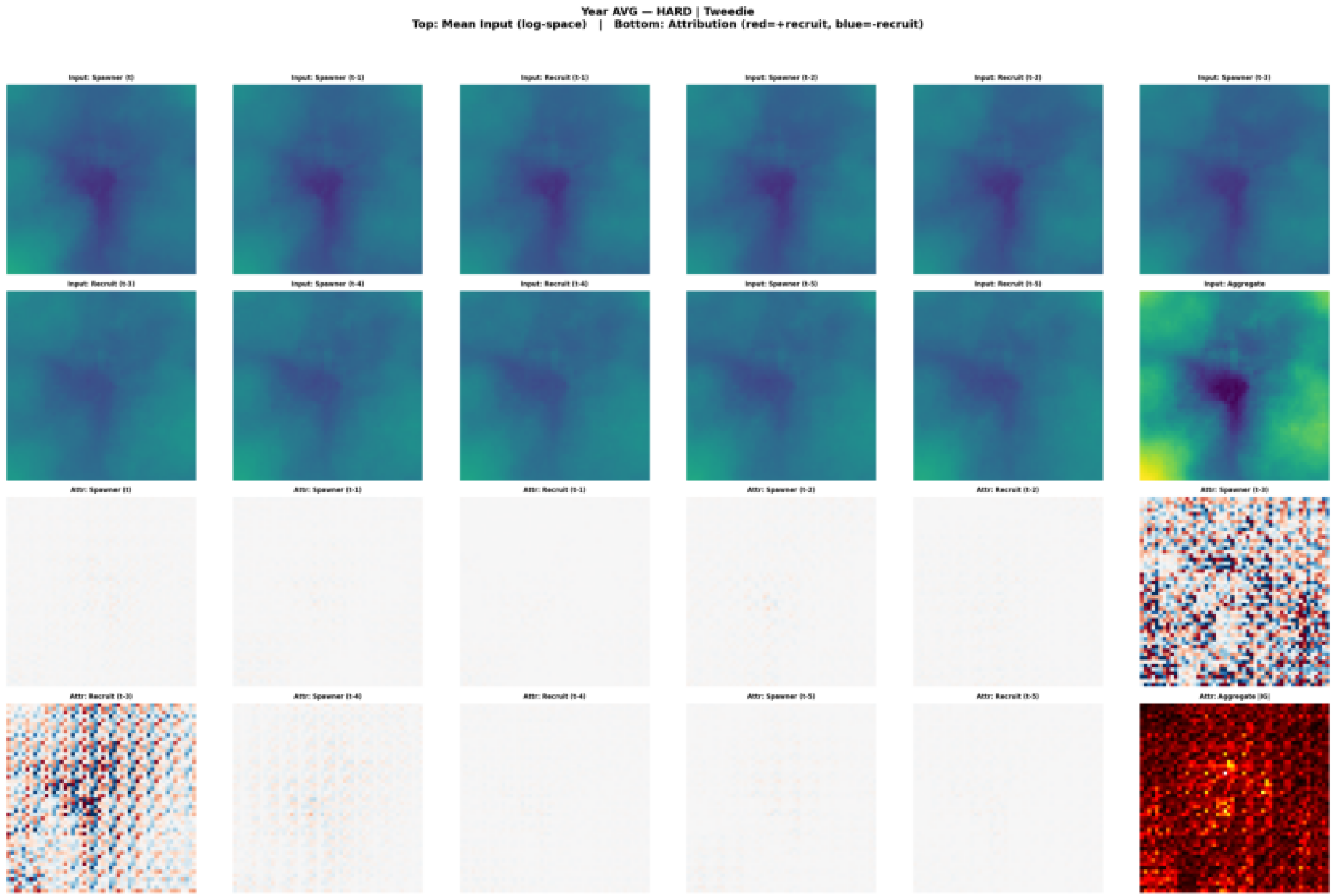


Annual Recruitment Deviations: HARD (Tweedie)
TRAIN
VAL
TEST
5th–95th
25th–75th
Median
RecDev
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
0
5
10
15
20
25
30

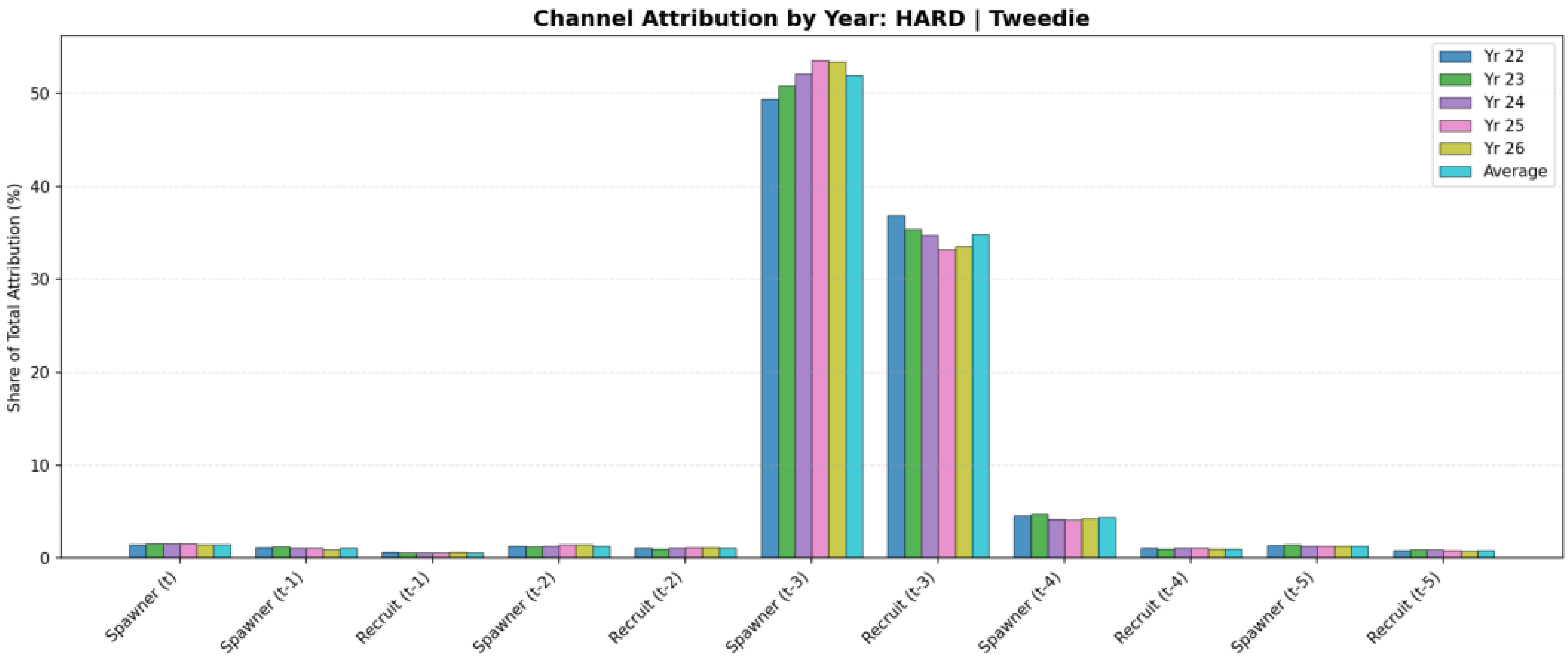


Annual Total Abundance (median, 25th–75th dark, 5th–95th light)
1e7
TRAIN
VAL
TEST
Target
Prediction
Spawner
Total Abundance
6
5
4
3
2
1
0
0
5
10
15
20
25
30

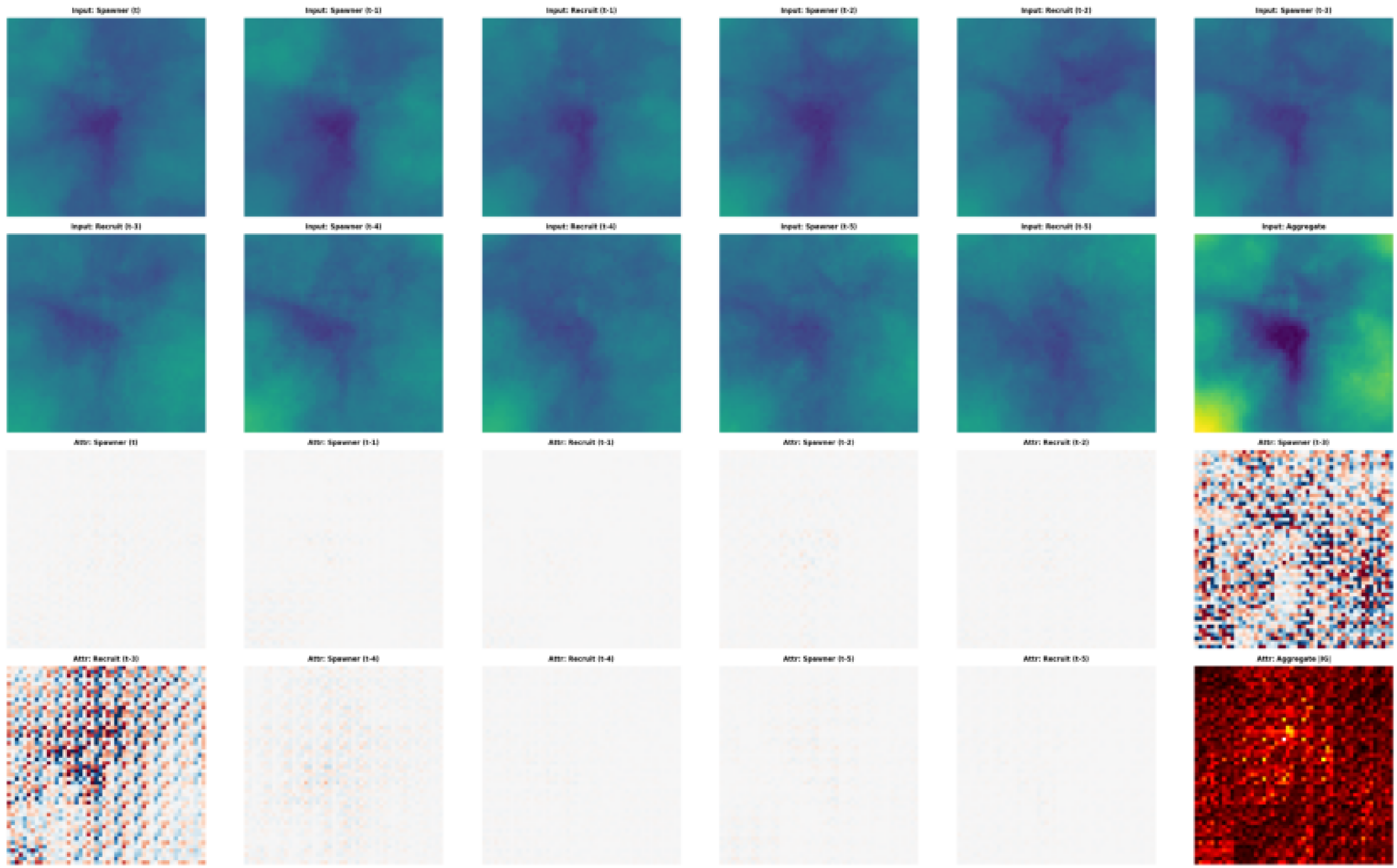
Spatial Trajectory: HARD | Tweedie

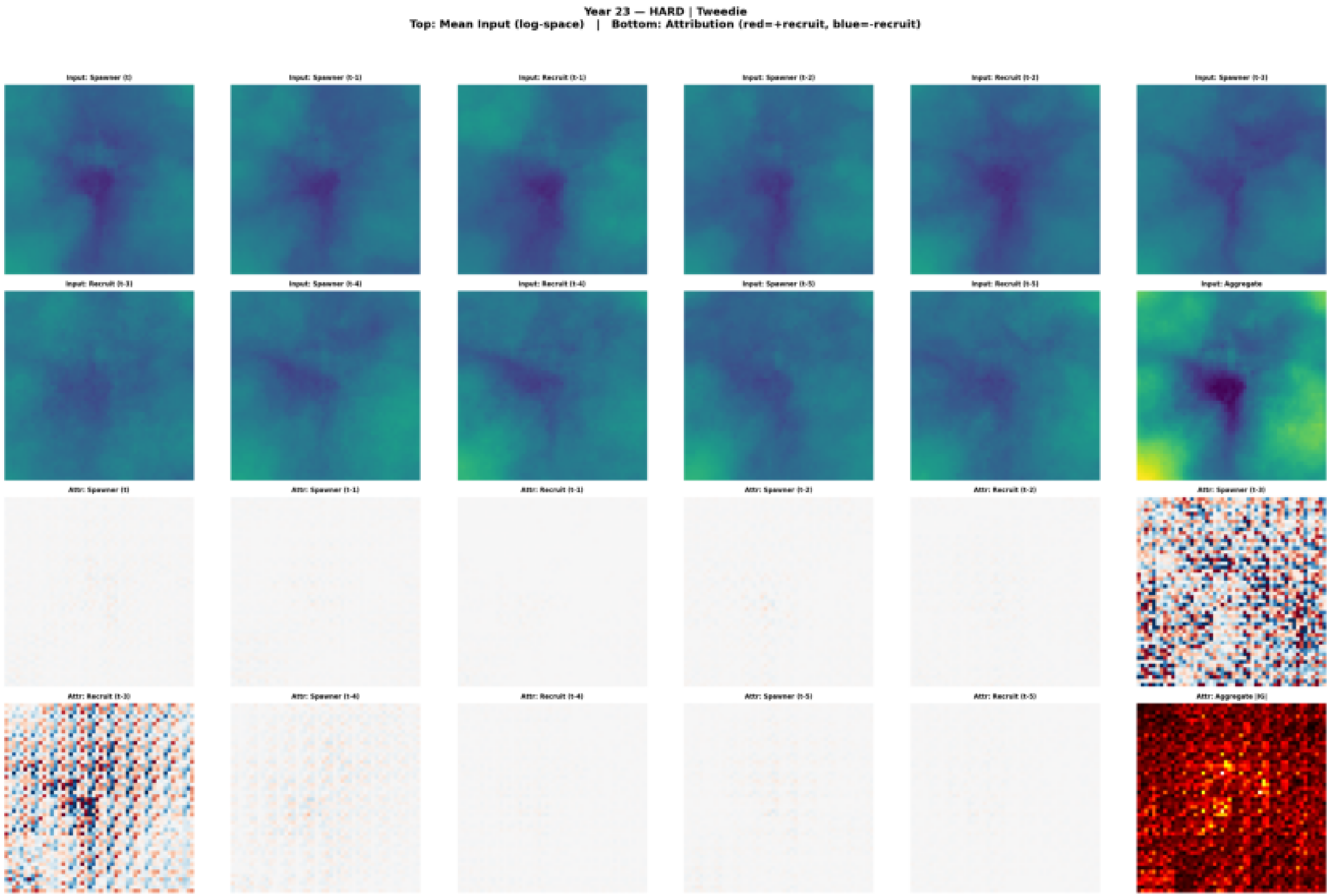


Year AVG — HARD | Tweedie
Top: Mean Input (log-space) | Bottom: Attribution (red=+recruit, blue=-recruit)
Input: Spawner (t)
Input: Spawner (t-1)
Input: Recruit (t-1)
Input: Spawner (t-2)
Input: Recruit (t-2)
Input: Spawner (t-3)
Input: Recruit (t-3)
Input: Spawner (t-4)
Input: Recruit (t-4)
Input: Spawner (t-5)
Input: Recruit (t-5)
Input: Aggregate
Attr: Spawner (t)
Attr: Spawner (t-1)
Attr: Recruit (t-1)
Attr: Spawner (t-2)
Attr: Recruit (t-2)
Attr: Spawner (t-3)
Attr: Recruit (t-3)
Attr: Spawner (t-4)
Attr: Recruit (t-4)
Attr: Spawner (t-5)
Attr: Recruit (t-5)
Attr: Aggregate (IG)

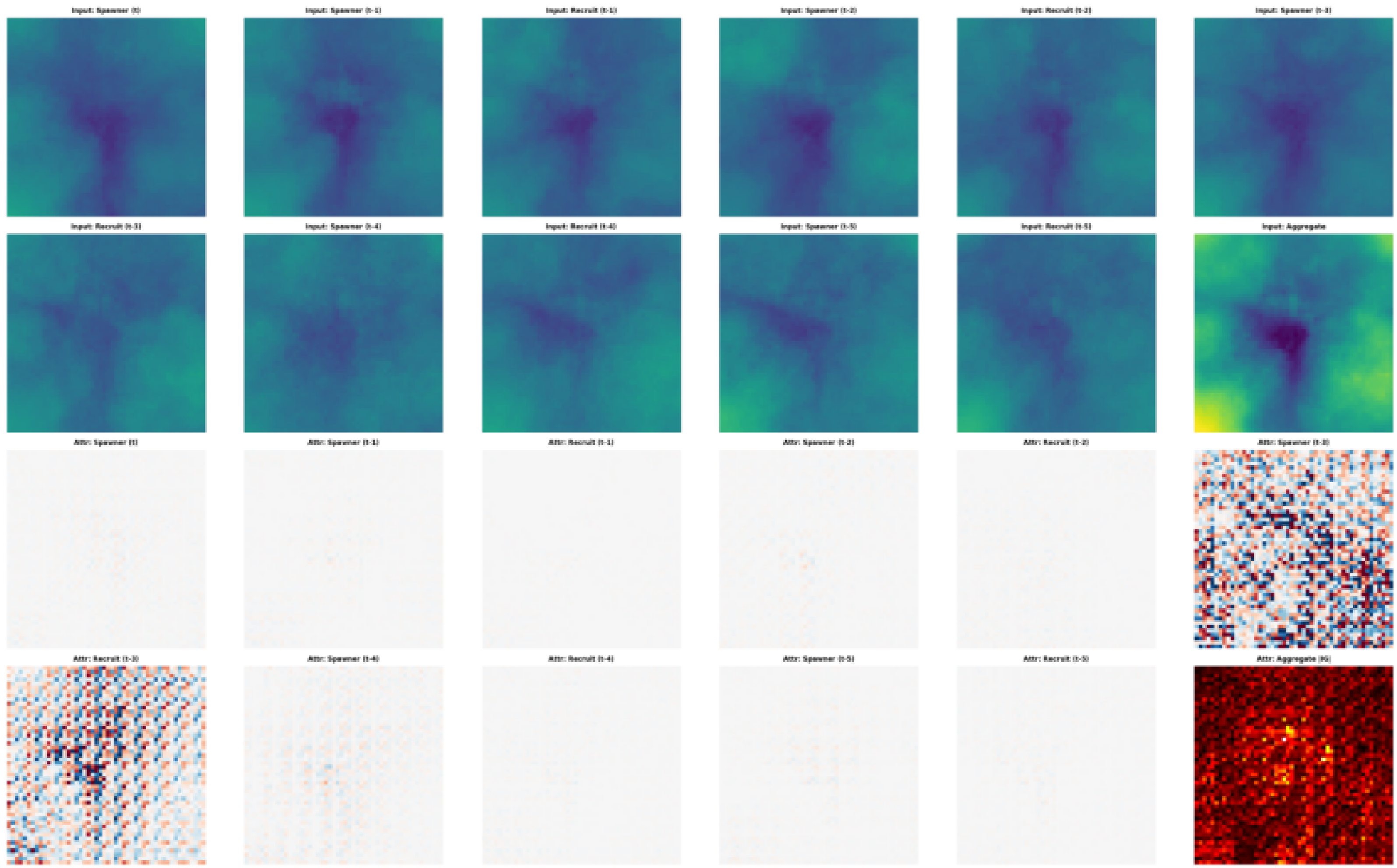

Channel Attribution by Year: HARD | Tweedie
Share of Total Attribution (%)
0
10
20
30
40
50
Yr 22
Yr 23
Yr 24
Yr 25
Yr 26
Average
Spawner (t)
Spawner (t-1)
Recruit (t-1)
Spawner (t-2)
Recruit (t-2)
Spawner (t-3)
Recruit (t-3)
Spawner (t-4)
Recruit (t-4)
Spawner (t-5)
Recruit (t-5)

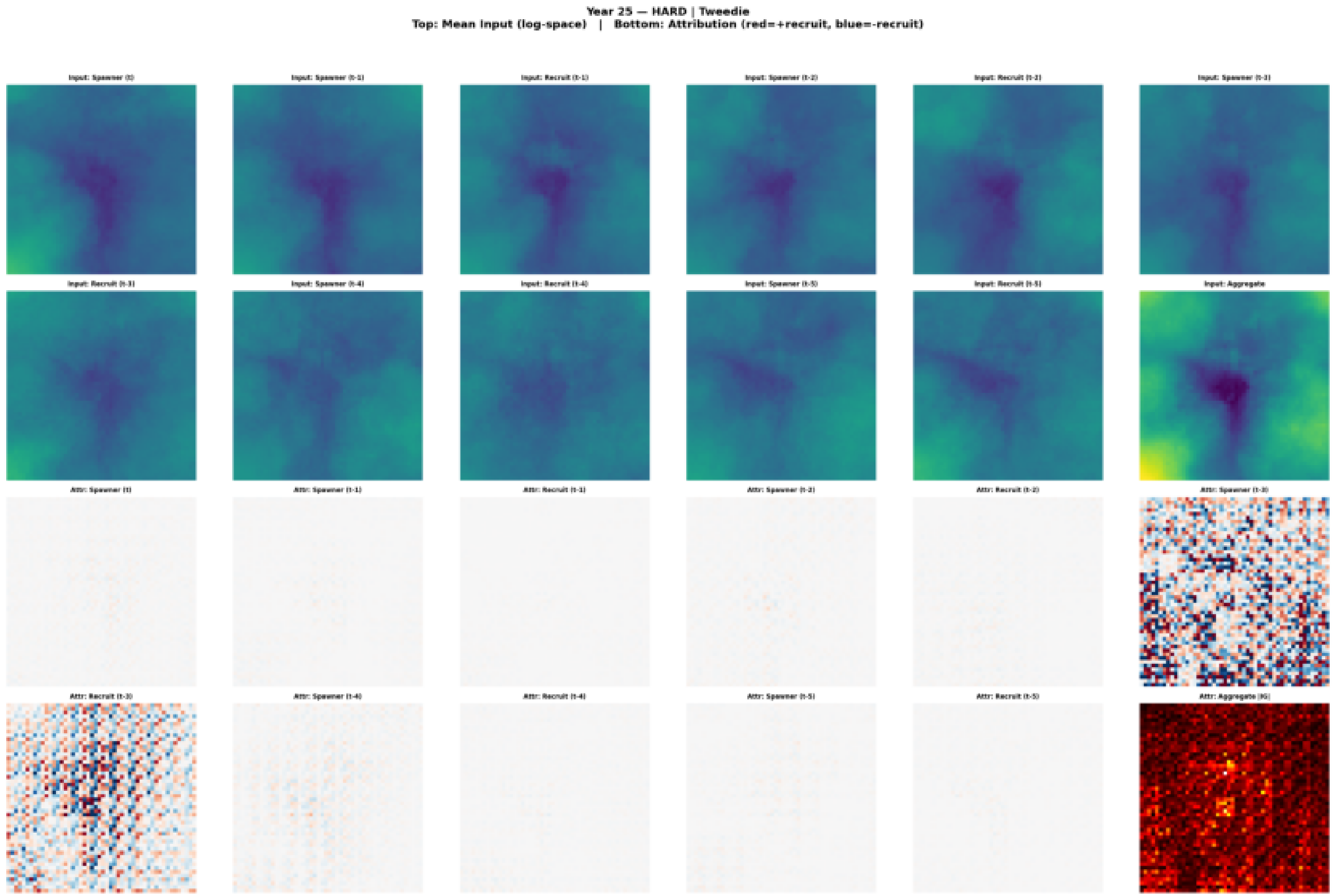
Year 22 — HARD | Tweedie
Top: Mean Input (log-space) | Bottom: Attribution (red=+recruit, blue=-recruit)




Year 23 — HARD | Tweedie
Top: Mean Input (log-space) | Bottom: Attribution (red=+recruit, blue=-recruit)
Input: Spawner (t)
Input: Spawner (t-1)
Input: Recruit (t-1)
Input: Spawner (t-2)
Input: Recruit (t-2)
Input: Spawner (t-3)
Input: Recruit (t-3)
Input: Spawner (t-4)
Input: Recruit (t-4)
Input: Spawner (t-5)
Input: Recruit (t-5)
Input: Aggregate
Attr: Spawner (t)
Attr: Spawner (t-1)
Attr: Recruit (t-1)
Attr: Spawner (t-2)
Attr: Recruit (t-2)
Attr: Spawner (t-3)
Attr: Recruit (t-3)
Attr: Spawner (t-4)
Attr: Recruit (t-4)
Attr: Spawner (t-5)
Attr: Recruit (t-5)
Attr: Aggregate (IG)

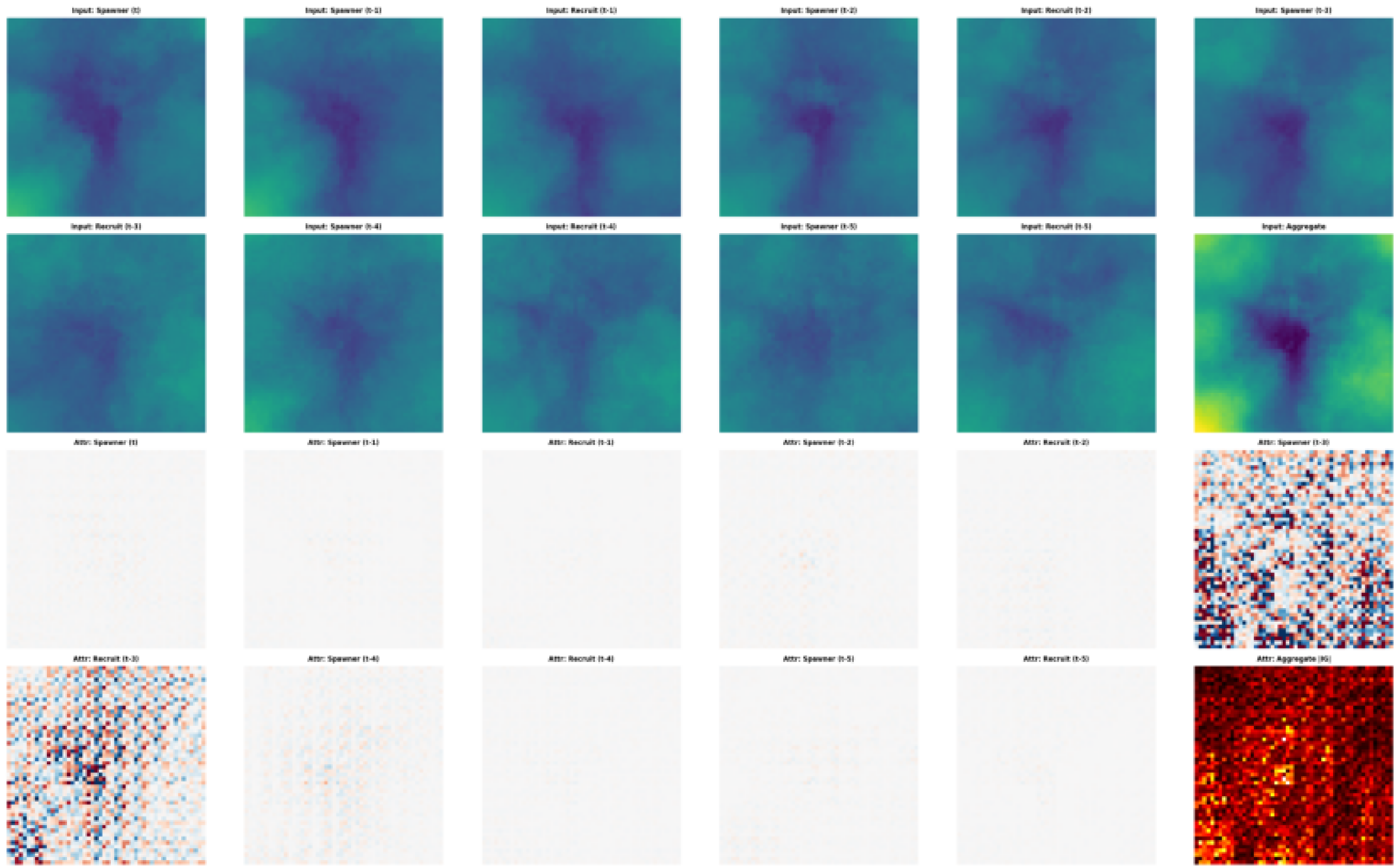
Year 24 — HARD | Tweedie
Top: Mean Input (log-space) | Bottom: Attribution (red=+recruit, blue=-recruit)





Year 25 — HARD | Tweedie
Top: Mean Input (log-space) | Bottom: Attribution (red=+recruit, blue=-recruit)
Input: Spawner (t)
Input: Spawner (t-1)
Input: Recruit (t-1)
Input: Spawner (t-2)
Input: Recruit (t-2)
Input: Spawner (t-3)
Input: Recruit (t-3)
Input: Spawner (t-4)
Input: Recruit (t-4)
Input: Spawner (t-5)
Input: Recruit (t-5)
Input: Aggregate
Attr: Spawner (t)
Attr: Spawner (t-1)
Attr: Recruit (t-1)
Attr: Spawner (t-2)
Attr: Recruit (t-2)
Attr: Spawner (t-3)
Attr: Recruit (t-3)
Attr: Spawner (t-4)
Attr: Recruit (t-4)
Attr: Spawner (t-5)
Attr: Recruit (t-5)
Attr: Aggregate |IG|

Year 26 — HARD | Tweedie
Top: Mean Input (log-space) | Bottom: Attribution (red=+recruit, blue=-recruit)



Summary Statistics: HARD | Tweedie

Phase	Pred_Mean	Tgt_Mean	Pred_SD	Pred_P50	Pred_P95	Abundance_Capture	MAE	Spearman
TRAIN	4692.9	4221.2	23726.8	591.5	15683.7	1.112	3983.5	0.648
VAL	3754.6	3598.1	14850.1	483.2	15029.1	1.043	3524.4	0.695
TEST	3443.7	3896.0	14316.5	426.2	12854.1	0.884	3417.5	0.700

Detailed Metrics (R-style): HARD | Tweedie

Phase	RMSE	RMSE_unbias	Bias	Bias_%	Spearman	Pearson	Pct_Error	Zero_F1	TopK_Acc	TopK_Jaccard	Quant_Err	Skill_MSE	Skill_MAE
TRAIN	2741.6	2674.3	472.2	4.8%	0.963	0.986	11.4%	0.000	0.859	0.756	0.55	0.900	0.747
VAL	1902.0	1873.5	157.3	3.2%	0.968	0.970	4.4%	0.000	0.806	0.678	0.53	0.908	0.722
TEST	2158.2	2084.1	-451.6	5.9%	0.960	0.980	-10.9%	0.000	0.840	0.725	0.58	0.911	0.738

Appendix A: Summary Metrics

APPENDIX A: Table Metric Definitions

All metrics computed on ORIGINAL DENSITY SCALE (after expm1 and bias correction for MSE). Zero threshold = 14 (smallest nonzero observed value): predictions AND observations below 14 are set to 0 before ALL metric computation. Both Page 7 and Page 8 apply this threshold.

Summary Table (Page 7)

Computed per bootstrap sample, then averaged across all samples in the phase.

Pred_Mean, Tgt_Mean
  Arithmetic mean of all cell values across all bootstraps (after thresholding).

Pred_SD
  Standard deviation of all cell values.

Pred_P50, Pred_P95
  Median and 95th percentile of all cells.

Abundance_Capture
  sum(pred) / sum(target) across all cells and bootstraps. 1.0 = perfect match.

MAE
  mean(|pred - target|) per sample on thresholded original-scale values,
  then averaged across samples.

Spearman
  Spearman rank correlation between flattened 50x50 pred and target grids.
  Computed per sample on thresholded original-scale values, then averaged.

Appendix A (cont): Detailed Metrics

```
APPENDIX A (cont): Detailed Metrics (Page 8)
════════════════════════════════════════════════════════════════

For Page 8, metrics are computed per-year by first AVERAGING the 50x50 spatial
field across all 100 bootstraps (both target and pred), then applying the zero
threshold, then computing metrics on that mean field. Per-year metrics are
averaged across years within each phase (TRAIN/VAL/TEST).

RMSE
  sqrt(mean((obs - pred)^2)) across all 2500 cells.

RMSE_unbiased
  RMSE after subtracting mean bias:
    corrected = pred - mean(pred - obs)
    RMSE_ub  = sqrt(mean((obs - corrected)^2))
  Isolates spatial pattern error from overall level error.

Bias
  mean(pred - obs). Positive = systematic overprediction.

Bias_%
  bias^2 / rmse^2 * 100
  High (>50%): main error is wrong overall level.
  Low: spatial pattern errors dominate.

Spearman
  Rank correlation on the thresholded, original-scale, bootstrap-averaged field.

Pearson
  Linear correlation on the same field. Sensitive to magnitude unlike Spearman.

Pct_Error
  (sum(pred) - sum(obs)) / sum(obs) * 100

Zero_F1
  F1 for zero-classification (threshold = 14):
    TP: obs < 14 AND pred < 14
    FP: obs >= 14 AND pred < 14
    FN: obs < 14 AND pred >= 14
    Prec = TP / (TP + FP),  Rec = TP / (TP + FN)
    F1   = 2 * Prec * Rec / (Prec + Rec)
```

Appendix A (cont): Ranking & Skill

```
APPENDIX A (cont): Detailed Metrics (Page 8)
═══════════════════════════════════════════════════════════════

TopK_Accuracy
  Top 10% of cells by obs vs pred density:
    k = ceil(0.1 * 2500) = 250 cells
    Acc = |obs_top250 & pred_top250| / 250
  Measures whether the model correctly identifies spatial hotspots.

TopK_Jaccard
  Jaccard index for the same top-k sets: |intersection| / |union|
  More conservative: penalizes both missed and false hotspots.

Quantile_Mean_Error
  Obs and pred independently binned into 10 quantile bins:
    - Cells < 14 get bin 0
    - Nonzero cells get bins 1-10 by quantile rank within nonzero values
  Metric = mean(|obs_bin - pred_bin|) across all 2500 cells.
  0 = perfect distributional match, 3 = cells are avg 3 bins off.
  NOTE: obs and pred use their OWN quantile edges, measuring distributional
  shape agreement, not just rank order.

Skill_MSE
  1 - MSE_model / MSE_climatology, where climatology = mean(obs) everywhere.
  Positive = beats the mean prediction. Negative = worse than the mean.
  Primary 'is the model useful' metric.

Skill_MAE
  1 - MAE_model / MAE_climatology. Less sensitive to outlier cells.

────────────── Difference between Page 7 and Page 8 ─────────────

Both apply the same zero threshold (14) on the same original density scale.
They differ in aggregation order:
  Page 7: compute metric per bootstrap, then average metrics
  Page 8: average bootstraps into mean field, then compute metric
Page 8 gives model performance on the expected spatial field (less noisy).
```

Appendix B: Plot Methodology

```
APPENDIX B: Plot Methodology
════════════════════════════════════════════════════════════════

── Page 2: Training Curves ────────────────────────────────────

Left: per-epoch train and val loss.
  MSE: loss in LOG-SPACE (on log(1+x) transformed values).
  Tweedie: loss on ORIGINAL SCALE (expm1 applied before loss function).
  Arrow marks epoch with lowest validation loss.

Right: OneCycleLR learning rate schedule (log scale).
  Ramps max_lr/10 -> max_lr (first 10%), then cosine-anneals to max_lr/1000.

── Page 3: Embedding Weight Magnitudes ────────────────────────

For each of 11 input channels (Spawner + 5 historical spawner-recruit pairs):
  importance = mean(|weights|) of channel's linear projection in PatchEmbedding.
  Min-max normalized to [0, 1] for comparison.

This reflects signal AMPLIFICATION at the embedding layer — a static property
of the trained weights, NOT causal importance or input-dependent attribution.
Compare with the IG attribution bar chart for the gradient-based view.

── Page 4: RecDev & Abundance ─────────────────────────────────

Top panel (RecDev):
  RecDev = (target_sum - pred_sum) / (target_sum + pred_sum) per sample.
  Red line = median across bootstraps. Ribbon = 5th to 95th percentile.
  Near 0 = balanced. Positive = underprediction. Negative = overprediction.

Bottom panel (Abundance):
  Median of grid-wide sums per year. Ribbons = 5th to 95th percentile.
  Target (black): sum(expm1(target_log))
  Pred (blue): sum(expm1(pred_log) * bias_correction)
  Spawner (green): sum(expm1(spawner_log))
```

Appendix B (cont): Spatial & Attribution

```
APPENDIX B (cont): Plot Methodology
═══════════════════════════════════════════════════════════════

── Page 5: Spatial Trajectory ─────────────────────────────────

Three columns per year: spawner input | true recruits | predicted recruits.
All displayed in LOG-SPACE with shared color scale (vmin=0, vmax=8).
Only the first bootstrap replicate is shown.

Prediction borders indicate phase:
  Green = TRAIN,  Orange = VAL,  Red = TEST

── Page 6: Integrated Gradients Attribution ───────────────────

Attribution computed via integrated gradients (Sundararajan et al. 2017).
For each validation year:

  1. Baseline = mean training input field (the 'boring' average-year reference)
  2. Interpolate from baseline to actual input in 50 steps (alpha: 0 -> 1)
  3. At each step, compute gradient of mean predicted recruitment w.r.t.
     the interpolated input: d(mean_pred) / d(input)
  4. Average gradients across all 50 steps
  5. Scale by (actual_input - baseline) to convert rate to total contribution

Result: attribution tensor [11, 50, 50] with same shape as input.
Each value = estimated contribution of that pixel/channel to mean predicted
recruitment. Averaged across bootstrap replicates per year to reveal stable
spatial patterns.

Top rows: mean input channels (log-space)
Bottom rows: attribution per channel (RdBu colormap)
  Red = increases recruitment prediction,  Blue = decreases
Aggregate panel = sum(|attribution|) across all 11 channels
```

Appendix B (cont): Bars & Corrections

APPENDIX B (cont): Plot Methodology

── IG Channel Bar Chart

Total |attribution| summed across all spatial cells for each of the 11 channels, expressed as percentage of total. Grouped by validation year plus grand average across years.

This is the gradient-based, input-dependent counterpart to the static embedding weight magnitudes on Page 3. Agreement between the two provides converging evidence for channel importance; disagreement indicates the transformer layers modulate channel usage beyond what embedding weights suggest.

── Phase Region Markers

All time-series plots include:

- Vertical dashed lines at year 21.5 (train/val) and 26.5 (val/test)
- Shaded backgrounds: green (train), orange (val), red (test)
- Text labels at top of each region

Year indices are 0-based:
  0-21 = TRAIN (22 years), 22-26 = VAL (5 years), 27-29 = TEST (3 years)

── Bias Correction (MSE only)

```
pred_original = expm1(pred_log) * exp(sigma^2 / 2)
```

where sigma^2 = var(target_log - pred_log) computed on the training set using the best checkpoint. Corrects Jensen's inequality: E[exp(X)] > exp(E[X]) when X has variance.

Tweedie: no correction needed — loss is computed on the original scale (expm1 is applied inside the forward pass before loss), so gradients naturally optimize for the correct conditional mean.